



Miriam Sarahi Lozano Gamboa

Molecular Biologist – Cancer Biology | Genomic Epidemiology | Molecular Diagnostics

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About Me

Molecular biologist with a strong foundation in cancer drug resistance, antimicrobial genomics, and molecular diagnostics. Skilled in NGS (RNA-Seq, DNA Prep, TruSeq RNA), PCR/qPCR, single-cell transcriptomics, and cell culture. Seeking a PhD opportunity in oncology, functional precision medicine, or host-pathogen interactions.

Education

Potosino Institute of Scientific and Technological Research (IPICyT),

San Luis Potosí, México

M.S., Molecular Biology, December 2020

- Thesis: *Diterpenes derived from Jatropha dioica modulate multidrug resistance in cell lines of human breast carcinoma*
- Advisor: Dr. Elihú Bautista
- Focus: *Cancer biology, chemotherapy resistance, cell signaling, bioinformatics, and in vitro assays*

Autonomous University of Aguascalientes (UAA),

Aguascalientes, México

B.S., Biotechnology, December 2018

- Thesis: *Micropropagation and determination of total phenolic compounds in Echinocactus spp*
- Major: *Genetic engineering and molecular techniques.*
- Advisor: Dr. Yenny Adriana Gómez Aguirre

Work Experience

Specialist in Molecular Diagnostics and Epidemiology, Health Institute of the State of Aguascalientes 2021 to present

Molecular biology department

- Managed high-throughput molecular workflows for genomic surveillance and clinical research (1,000+ samples/year).
- Designed and validated qPCR assays for detection of pathogens and resistance markers.
- Conducted NGS library preparation (RNA/DNA), data generation, and reporting for MDR bacterial isolates

Microbiology Technician, Health Institute of the State of Aguascalientes

2021 to 2022

Department of Microbiology

- Standardized workflows for bacterial cultures, antibiograms, and biosafety protocols in clinical isolates.

Research Assistant, Autonomous University of Ags (UAA)

2015 to 2018

Department of Plant Biotechnology

- Optimized *in vitro* propagation techniques and quantified bioactive compounds for biotechnological applications.

Research Internships

Establishment of Chemotherapy-Resistant Breast Cancer Cell Lines, National Autonomous University of Mexico (UNAM) 2019

Molecular biology department

- Established vinblastine-resistant breast cancer cell lines for functional assays.
- Investigated molecular mechanisms of resistance and metastatic behavior.
- Advisor: **Dr. Mabel Clara Fragoso Serrano**

Neuroblastoma Cell Lines and Animal Models, Autonomous University of Ags (UAA) 2017

Department of Physiology and Pharmacology

- Acquired *in vivo* and *in vitro* model handling skills using neuroblastoma systems.
- Advisor: **Dr. Andrés Quintanar-Stephano**

Grants and Fellowships

- **Research Stay Scholarship, IPICYT / UNAM** *PI: Dr. Rogelio Pereda-Miranda*
Project: Regulation of chemotherapy resistance in breast cancer cell lines
- **CONACYT Scholarship – Master’s Degree** (Fully funded)
- **CONACYT Scholarship – Undergraduate Research** *PI: Dr. Yenny A. Gómez Aguirre*
Project: Propagation and rational use of drought-adapted plants
- **I-MMAS Research Grant, University of Illinois Program** *PI: Dr. Corina-Diana Ceapă*
Project: Genomic library of antibiotic-resistant bacteria isolated in Mexico
- **Erasmus+ Summer Scholarship, Vive México / European Union** *Germany, 2024*
- **SMBE Travel Grant, INTERMET S.A. DE C.V.** 2024

Teaching Experience

Lecturer – Molecular Biology, Cuauhtémoc University, Aguascalientes

2022-to present

Course for Medical Program (30+ students per class)

Taught core molecular biology concepts with emphasis on laboratory techniques and clinical relevance.

Mentoring Undergraduate Biotechnology Thesis, Autonomous University of Ags (UAA) 2022 to 2024

Supervised a student thesis on genetic variants in *Escherichia coli* from clinical isolates. Provided training in bioinformatics tools and molecular techniques.

Volunteering

Service Civil International (SCI), Marth, Germany

2024

Participated in ecological farming and sustainability projects. Promoted conservation and intercultural collaboration with international volunteers.

Manuscripts in Preparation, *drafts available on request*

- 1.1 **Lozano Gamboa, M. S.**, Hernández Magro Gil, K. G., Muñoz Gomez, B., Palacios Marmolejo, A., & Ceapă, C. D. (2024). Multi-Drug-Resistant *Klebsiella Spp.* And *Escherichia Coli* Isolated From Central Mexico Between 2020 And 2023, A Genetic Analysis. Submitted to The Lancet Microbe (2025).
- 1.2 Alejo, M. A., Hernández Magro Gil, K. G., Villavicencio, P., Muñoz Gomez, B., **Lozano Gamboa, M. S.**, Palacios Marmolejo, A., Figueroa, M., García-Contreras, R., & Ceapă, C. D. Resistome Exploration Of Post-Covid Multi-Drug Resistant *Acinetobacter Baumannii* Isolated From Mexico. Submitted to The Lancet Microbe (2025).
- 1.3 Ceapă, C. D., **Lozano Gamboa, M. S.**, Hernández Magro Gil, K. G., Muñoz Gomez, B. & Palacios Marmolejo, A., Comparative Genomics of *Pseudomonas aeruginosa* Isolates From Mexico.
- 3.1 **Lozano Gamboa, M. S.** (2024). Terpenes of Plant Origin and Their Modulating Activity on Multidrug Resistance in Human Breast Cancer Cell Lines: A Review Article. *Zenodo*. DOI: [10.5281/zenodo.11513941](https://doi.org/10.5281/zenodo.11513941).

Publications in Peer-Review

- 2.1 Elihú Bautista, **Sarahí Lozano-Gamboa**, Mabel Fragosó-Serrano, José Rivera-Chávez, Luis A. Salazar-Olivo, Jatrophenediol, a pseudoguaiane sesquiterpenoid from *Jatropha dioica* rhizomes, Tetrahedron Letters, 2022, 154040, ISSN 0040-4039, <https://doi.org/10.1016/j.tetlet.2022.154040>.
- 2.2 Muñoz Gomez, B., **Lozano Gamboa, M. S.**, Ceapa, C. D., & Palacios Marmolejo, A. (2024). Mortality Associated Sars-Cov2 Genomic Variants From Patients Hospitalized For Severe Pneumonia In Aguascalientes, Mexico From 2020 To 2023. Biorxiv, 2024.03.05.583594. DOI: <https://doi.org/10.1101/2024.03.05.583594>. Under revision at The molecular phylogenetics and evolution journal.

Conference talks and posters

- 4.1 **Lozano Gamboa, M. S.**, Hernández Magro Gil, K. G., Muñoz Gomez, B., Palacios Marmolejo, A., & Ceapă, C. D. (2024) Genomic Insights into the Evolutionary Trajectory of Antibiotic Resistance in *Klebsiella spp.* and *Escherichia coli*: A Study from Central Mexico (2020-2023) . SMBE Society for Molecular Biology and Evolution International Annual Meeting, Jalisco, Mexico. ([Watch poster here](#)).
- 4.2 **Lozano Gamboa, M. S.**, Muñoz Gomez, B., Ceapa, C. D., & Palacios Marmolejo, A. (2024). Mortality Associated Sars-Cov2 Genomic Variants. XLVIII National Congress of Infectology and Clinical Microbiology , Puebla, Mexico. ([Watch poster here](#)).
- 4.3 **Lozano Gamboa, M. S.**, Pérez Molphe Balch E. M., Gómez Aguirre. Y. A. Micropropagation and determination of total phenolic compounds in *Echinocactus spp* (2017) 34th IOS Congress, Querétaro, Mexico.

Additional Courses

Bioinformatics and Computational Analysis:

- Data Science with R and RStudio (Udemy, 2024)
- Machine Learning and Data Science: Complete Course with Python (Udemy, 2023)
- LINUX and PYTHON in Bioinformatics (Pharmaceutical and Biotechnological Innovation-Services SAS de CV, 2022)

Technical Skills:

- Update on Nucleic Acid Extraction, qPCR Design, and Standardization (ITRASIG, 2023)

Cancer and Molecular Biology:

- Epigenetic Control of Gene Expression, The University of Melbourne (Coursera, 2024)
- Cancer Biology Specialization, Johns Hopkins University (Coursera, 2024)
- Molecular Immunology (Autonomous University of Aguascalientes UAA, 2023)

Data Analysis and Research Skills:

- Excel for Intermediate and Advanced Students (Cuauhtémoc University, 2023)
- Scientific Writing (UNAM, 2023)

Technical Skills

- **PCR Techniques:** Experienced in PCR and qPCR design, optimization, and standardization for molecular diagnostics and research.
- **Next-Generation Sequencing:** Expertise in Illumina platforms (CovidSeq, RPIIP, DNA Prep, RNA Prep, TruSeq RNA, 16S), DNA/RNA library preparation, and single-cell techniques (scRNA-seq, scATAC-seq).
- **Cell Culture and Western Blotting:** Skilled in cell culture techniques and Western blotting

Languages

- Spanish: Native
- English: Advanced (TOEFL IBT score 96 2023)
- German: Basic (currently learning)

Online Portfolio

For more details please visit: <https://miriamsarahilozanogamboa.github.io/mmsarahi-lzgamboa/>